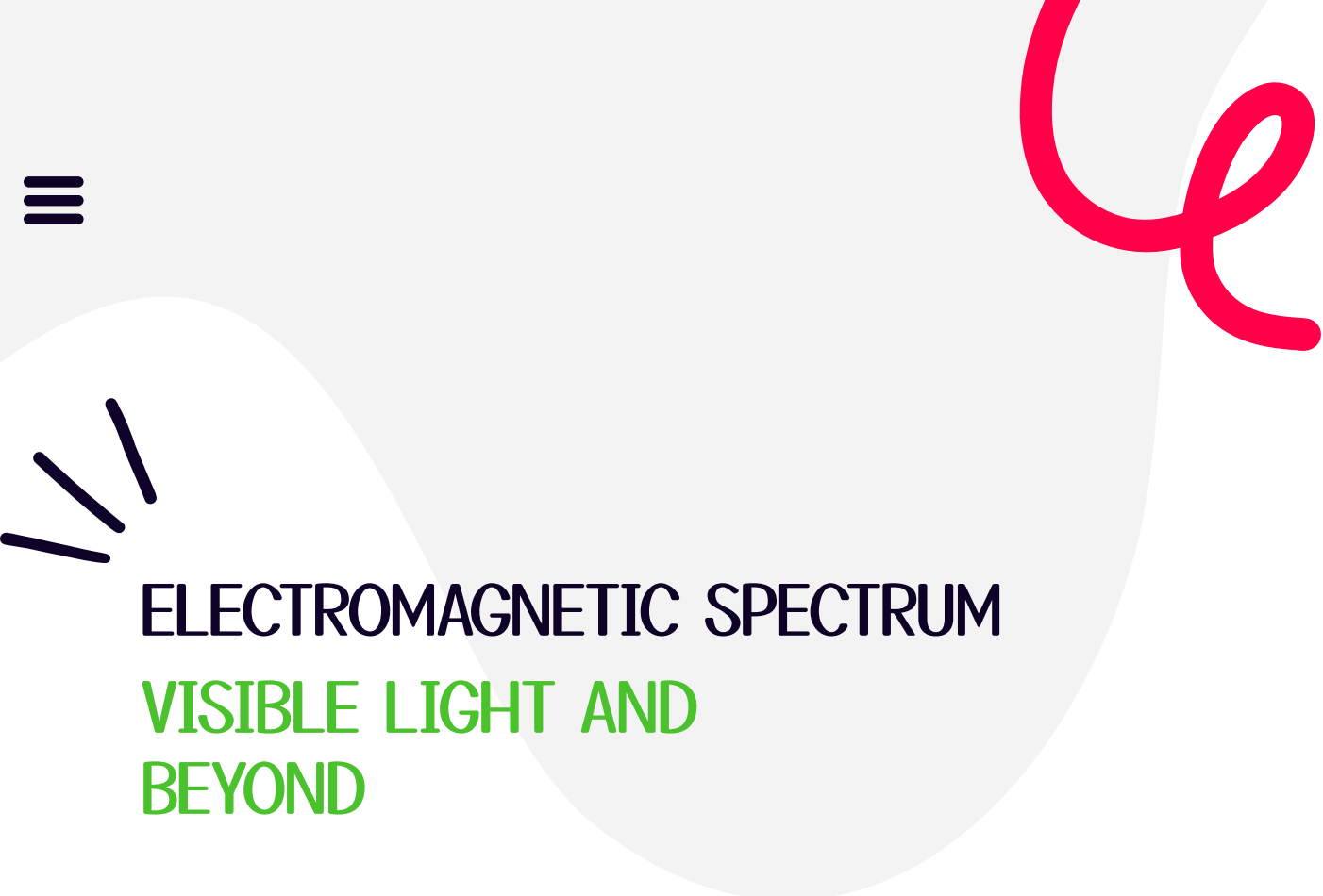




## ELECTROMAGNETIC SPECTRUM VISIBLE LIGHT AND BEYOND

Get ready to explore the amazing world of light and invisible waves! From the colorful rainbow to Wi-Fi signals, discover how electromagnetic waves are everywhere around us.



SEE THE RAINBOW COLORS ALL AROUND US



LEARN ABOUT INVISIBLE WAVES LIKE WI-FI



DISCOVER THE  
COLORFUL  
WORLD OF LIGHT  
WAVES!

LET'S BEGIN!

01

02

03





# CAN YOU SEE WI-FI?



—————→ **THINK ABOUT INVISIBLE THINGS AROUND YOU!**

Look around you! Wi-Fi helps you watch videos. TV remotes change channels. X-rays help doctors see your bones. Rainbows appear after rain. These all use special waves that travel through the air!

**LET'S EXPLORE TOGETHER!**

**WHAT DO THESE  
HAVE IN COMMON?**

TV remotes, Wi-Fi signals, X-ray machines, and rainbows all use waves from the electromagnetic spectrum. Some we can see, but most are invisible!

**INVISIBLE WAVES ARE ALL AROUND US!**



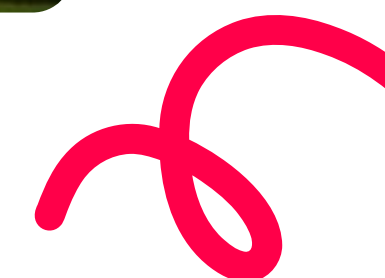


# OUR MONSOON GIFT – THE RAINBOW!

When sunlight passes through raindrops during monsoon, it splits into 7 beautiful colors! This magical arc in the sky is called Indradhanush . Each raindrop acts like a tiny prism, bending light into its colorful parts.

**7** COLORS IN EVERY RAINBOW — REMEMBER ROYGBIV!

**1** MNEMONIC: RICHARD OF YORK GAVE BATTLE IN VAIN





# ROYGBIV COLORS

## RAINBOW!



### RED - Orange - Yellow

The warm colors of sunrise and marigolds! Red has the longest wavelength in visible light.

### GREEN - Blue - Indigo

Cool colors like peacock feathers! Hindi Mnemonic: "Raju Orange "Friend turned blue because of Violet in Indore"

### VIOLET - Shortest wavelength

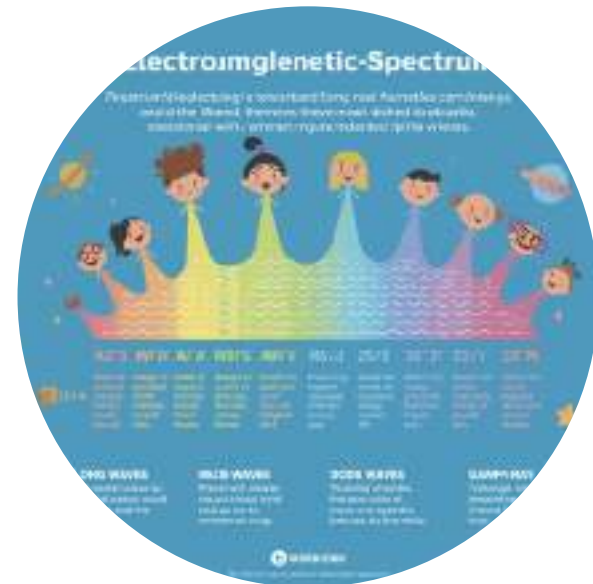
Remember these colors!  
English: "Richard Of York Gave Battle In Vain"





# THE EM SPECTRUM FAMILY

“LIKE A MEAL SERVED ON A PLATTER!”



RADIO & MICROWAVES

INFRARED & VISIBLE LIGHT

UV, X-RAY & GAMMA

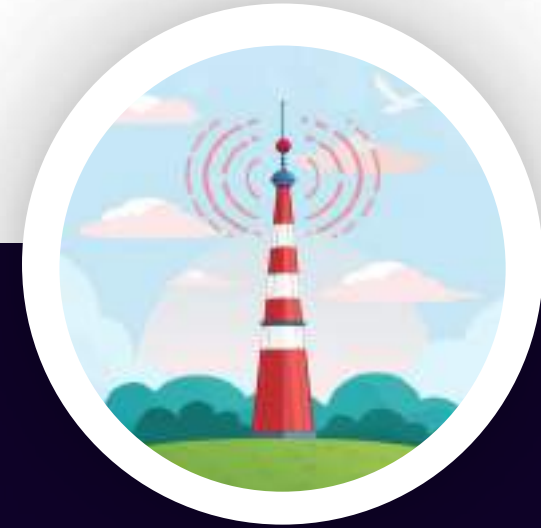


The Electromagnetic Spectrum is like a Thali with 7 dishes (waves)! Just like each dish has different flavors, each wave type has different energy and uses. All waves travel at light speed!

From longest Radio waves to shortest Gamma rays, the spectrum includes: Radio, Microwave, Infrared, Visible Light, Ultraviolet, X-Ray, and Gamma Rays. Each has special powers!



# RADIO WAVES – THE LONGEST!



**LONGEST  
WAVELENGTH, LOWEST  
ENERGY WAVES!**

→ **WHERE DO WE FIND RADIO WAVES?**

**Indian Examples:**

- All India Radio (Akashvani) - brings music and news to every home
- Doordarshan TV - entertainment for families
- Wi-Fi routers - internet at home and school
- Mobile networks (Jio, Airtel) - connecting all of India!

**RADIO WAVES TRAVEL SUPER FAR!**

Radio waves are the gentle giants of the EM spectrum! They have the longest wavelength and carry the least energy. That's why they're safe to use everywhere around us, from broadcasting to communication.

**FROM AKASHVANI TO YOUR WI-FI ROUTER!**





# MICROWAVES

## COOKING & COMMUNICATION



### COOKING & COMMUNICATION WAVES



#### SHORTER WAVES, MORE ENERGY!

Did you know your microwave oven uses the same type of waves that help satellites talk to Earth? Microwaves can heat food by making water molecules vibrate really fast! That's how your dal makhani gets hot in minutes. These waves are also used by ISRO satellites orbiting above India.

Microwaves have slightly shorter wavelengths than radio waves but carry more energy! ISRO uses them for satellite communication, and IMD weather radar tracks monsoon clouds using microwave technology.

**FROM HEATING BUTTERY LENTILS TO TRACKING MONSOONS!**





# INFRARED HEAT RAYS WE FEEL



—————→ **INVISIBLE WAVES CARRYING HEAT ENERGY**

Infrared waves are invisible heat rays that sit just below red light in the spectrum. We cannot see them with our eyes, but we can feel them as warmth! When you stand near a hot tandoor or feel the sun's warmth on your skin, that's infrared radiation at work.

**FEEL THE HEAT YOU  
CANNOT SEE!**



TV remote controls send infrared signals to change channels. The hot tandoor oven gives off infrared heat to cook naan and roti. COVID thermometers measure your forehead temperature using infrared. ISRO satellites use infrared to study Earth's weather!

**INFRARED WAVES ARE INVISIBLE HELPERS IN OUR DAILY LIFE!**







# VISIBLE LIGHT

## THE COLORS WE SEE!

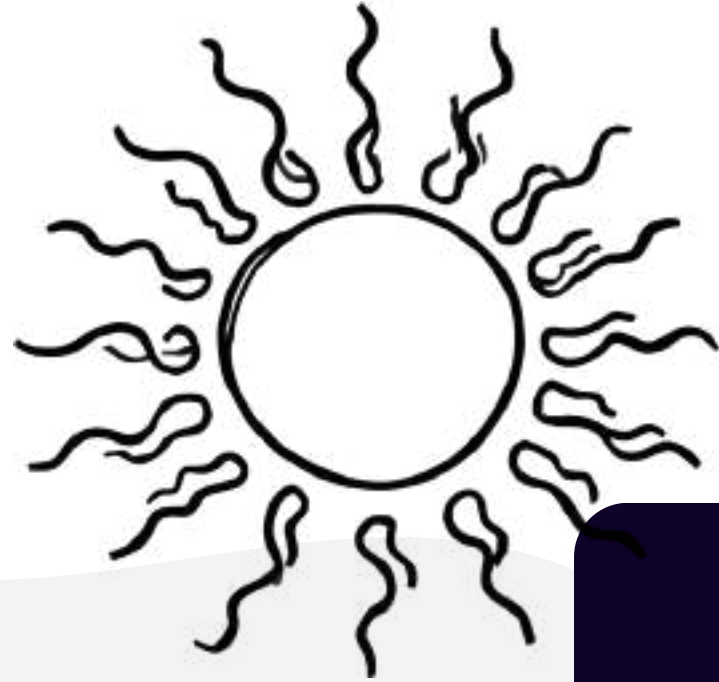


Visible light is the **ONLY** part of the electromagnetic spectrum our eyes can see! It gives us all the beautiful colors around us — from the rainbow after monsoon rains to the bright colors of Holi gula!



Look around you! Traffic lights use visible light to keep us safe. Festival rangoli uses all the rainbow colors. Every color you see is visible light bouncing into your eyes!





# ULTRAVIOLET BEYOND VIOLET

—————→ **INVISIBLE RAYS WITH SURPRISING POWERS!**

UV rays from the sun can cause sunburn after playing cricket outside. But morning sunlight helps our body make Vitamin D which keeps our bones strong! That's why a little sunshine is good for you.

**DID YOU KNOW? UV MEANS "BEYOND VIOLET"!**



## HIGHER ENERGY WAVES BEYOND VIOLET!

Bees can see UV patterns on flowers that we cannot!  
Indian banknotes have special UV security markings.  
Hospital UV lamps kill germs and keep us safe.

**UV WAVES ARE INVISIBLE BUT POWERFUL — THEY HELP AND CAN HARM!**





# X-RAYS

## POWERFUL WAVES



**VERY SHORT WAVELENGTH, VERY HIGH ENERGY!**

**Where do we use X-rays?**

- Hospital X-rays to check broken bones
- Airport security scanners to see inside bags
- ISRO & DRDO use them to check rocket parts for cracks

**Fun Fact: X-rays were discovered by accident in 1895!**

**X-RAYS CAN SEE THROUGH THINGS!**

**SEEING INSIDE OUR BODIES!**



X-rays have very short wavelengths and high energy. They can pass through soft things like skin but stop at hard things like bones. That's why doctors use them to see broken bones inside you!

**X-RAYS HELP DOCTORS, AIRPORTS & EVEN ISRO!**





# GAMMA RAYS

## SUPER POWERFUL!



**TINY WAVES, HUGE ENERGY!**

### Key Facts:

- Shortest wavelength of all EM waves
- Highest energy - very powerful!
- Can pass through almost anything
- Used in cancer treatment machines
- BARC Trombay studies nuclear science
- ISRO AstroSat finds gamma rays in space

**GAMMA RAYS HELP DOCTORS FIGHT DISEASES!**



**THE STRONGEST  
WAVES IN THE  
UNIVERSE!**



Gamma rays have the shortest wavelength and highest energy of all EM waves! They're used at AIIMS and Tata Memorial Hospital to treat cancer. BARC Trombay uses them to sterilize medical equipment. ISRO's AstroSat telescope detects gamma rays from space!

**FROM FIGHTING CANCER TO EXPLORING SPACE!**



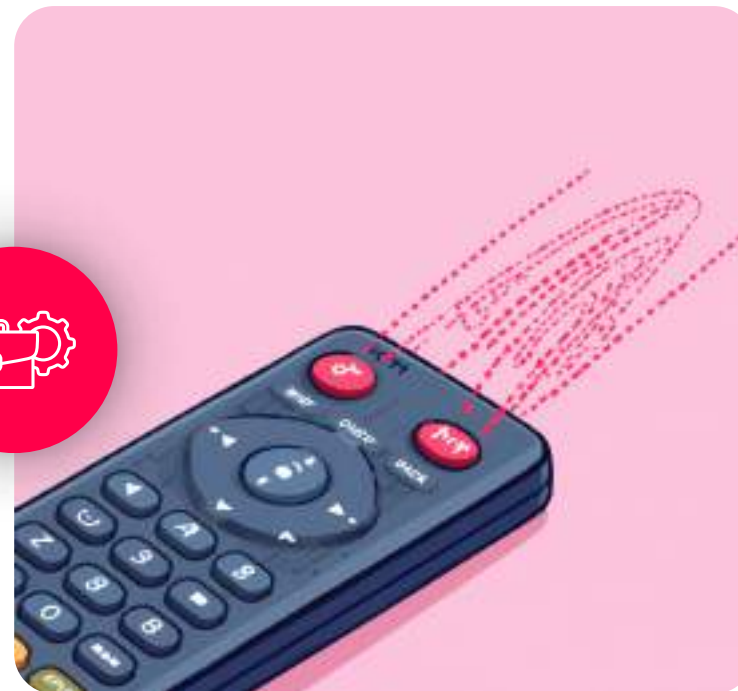
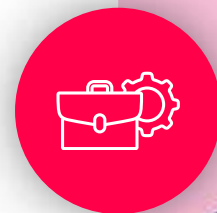




# HOW DOES A TV REMOTE WORK?



Infrared waves are special heat rays that we cannot see with our eyes. They travel in straight lines from the remote to the TV. That's why you need to point the remote at the TV for it to work!



**DEEP DIVE: THE  
MAGIC OF INFRARED  
SIGNALS**

When you press a button on your TV remote, it sends out invisible infrared light waves. These waves travel through the air and reach a sensor on your TV. The TV reads the signal and follows your command — like changing channels or volume!

**INVISIBLE HEAT WAVES CARRY YOUR COMMAND!**





## WHY MUMMY SAYS NO AFTERNOON SUN!



UV waves from the hot afternoon sun can burn your skin! But gentle morning sunlight helps your body make Vitamin D, which keeps your bones strong and healthy.



**UV WAVES: SUNBURN  
& BEE VISION**

Bees have special eyes that can see UV patterns on flowers that we cannot see! These patterns guide bees to nectar, like landing strips at an airport. Nature is amazing!

**INVISIBLE HEAT WAVES CARRY SECRETS!**





# THE BROKEN BONE STORY



Have you ever wondered how doctors can see inside your body without cutting it open? X-ray machines send invisible waves through you. Bones appear white because they stop the rays, while soft tissue looks darker. It's like shining a torch behind your hand!



HOW DOCTORS SEE  
YOUR BONES  
WITHOUT SURGERY!

X-rays are like magical light that can see through your skin! When doctors need to check if a bone is broken, they use X-rays. The rays pass easily through soft parts like skin and muscles, but bones are hard and block them. This creates a shadow picture showing your bones clearly!

**X-RAYS HELP DOCTORS HEAL US!**







# EM WAVE SORTING GAME!



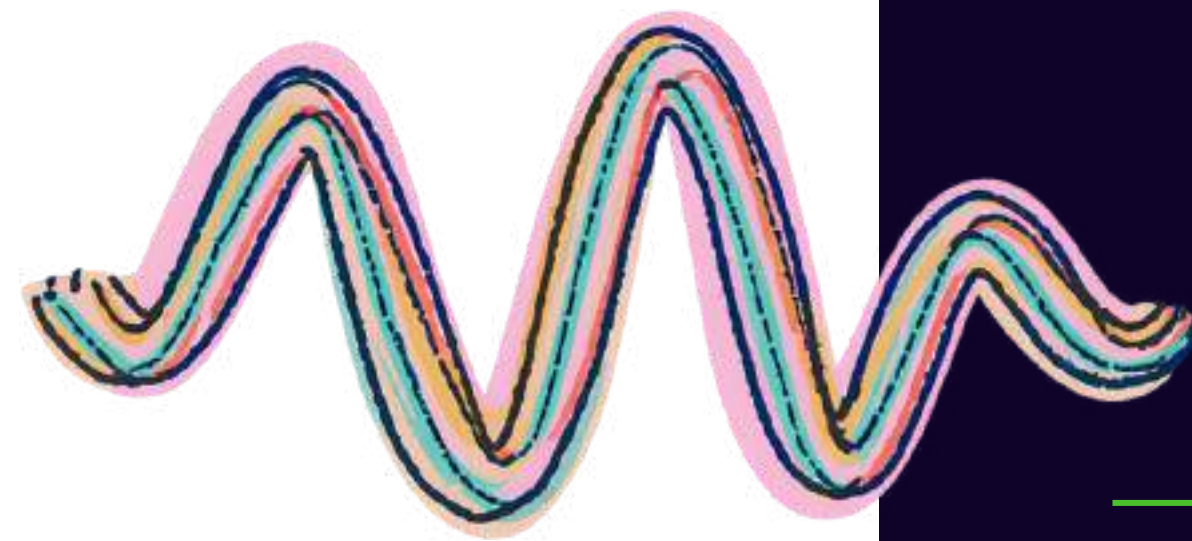
RADIO WAVES



VISIBLE LIGHT



INFRARED WAVES

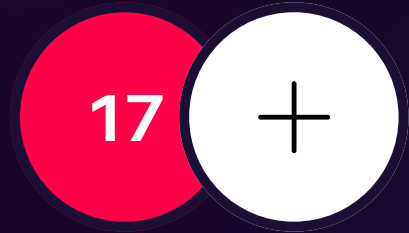


Match the picture to the EM wave type! Look at each image below and guess which electromagnetic wave it uses. Is it Radio, Microwave, Infrared, Visible, UV, or X-Ray?

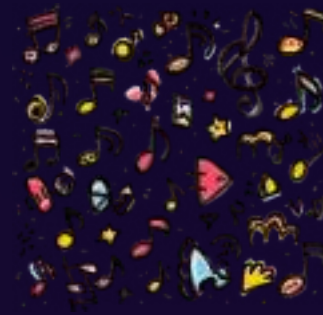
Hints: Doordarshan antenna uses Radio waves, X-ray hand uses X-Rays, monsoon rainbow is Visible light, TV remote uses Infrared, sunburn is from UV rays, microwave oven uses Microwaves!



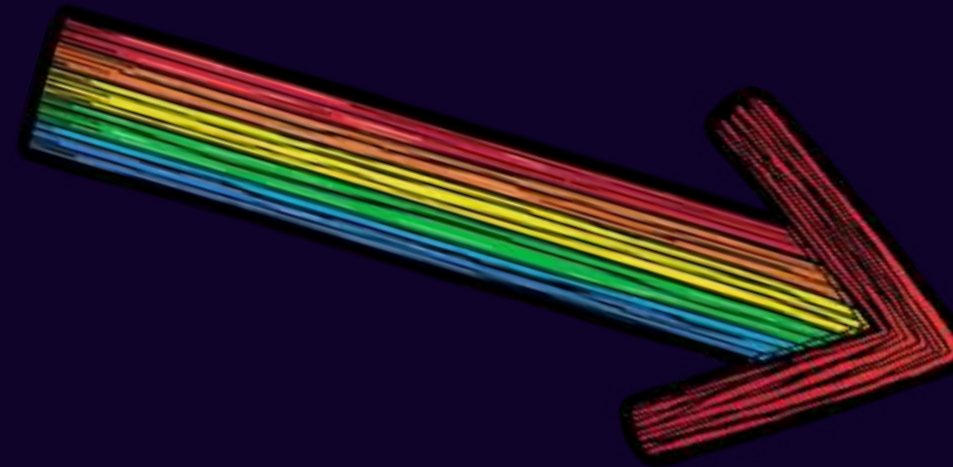




# EM SPECTRUM



## CHANT!



RADIO → MICRO → INFRA → VISIBLE!

ULTRA → X-RAY → GAMMA!

SAY IT LOUD WITH HAND GESTURES!



## RAPID FIRE QUIZ

# RAPID FIRE QUIZ TEST YOURSELF!



**Q1: TV REMOTES?**  
**ANSWER: INFRARED**



**Q2: SEE BONES?**  
**ANSWER: X-RAYS**



**Q3: SUNBURN?**  
**ANSWER: ULTRAVIOLET**



Q4: Which wave has the MOST energy?

Answer: Gamma Rays!

Q5: Name the 7 colors of visible light.

Answer: ROYGBIV - Red, Orange, Yellow, Green, Blue, Indigo, Violet!

Quick Tip: Remember these answers! The EM spectrum goes from lowest energy (Radio) to highest energy (Gamma Rays). Visible light is right in the middle!



# TODAY'S BIG IDEA

## SUMMARY

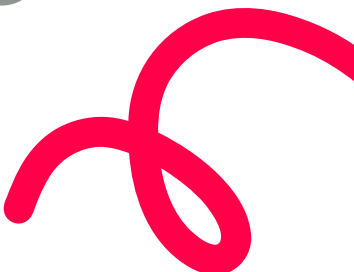
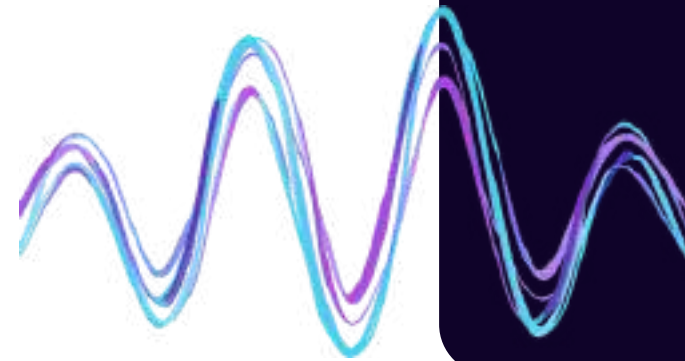
Visible light is just ONE member of a huge invisible family called the Electromagnetic Spectrum. Invisible waves like Wi-Fi and X-rays work silently all around us every day! From the radio waves bringing your favorite songs to the gamma rays helping doctors treat patients, these invisible helpers are everywhere.

7

TYPES OF WAVES IN THE EM SPECTRUM FAMILY



INVISIBLE WAVES WORKING AROUND YOU RIGHT NOW!







# HOMEWORK MISSION TIME!



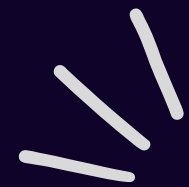
Go home and find 5 things that use invisible EM waves! Look for: TV remote, Wi-Fi router, microwave oven, mobile phone, radio.



Be a wave detective! Draw or write about each item you find. Which EM wave does it use? Bring your findings to share with the class!







# THANK YOU!



Any questions? We hope you enjoyed learning about the amazing Electromagnetic Spectrum today! Remember, invisible waves like Wi-Fi, X-rays, and infrared are working all around us every day. Keep exploring and stay curious!

